CRISPR in Precision Medicine: Transforming Gene Editing with Artificial Intelligence

Rushmila Islam
Rushmila.Islam@gmail.com

1 Introduction

Healthcare is moving away from a one-size-fits-all model toward precision medicine. Precision medicine aims to tailor medical treatment and interventions to the individual characteristics of each patient, taking into account variability in genes, environment, and lifestyle. At the core of this revolution is Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) based genome editing, a technology that has given researchers a direct, programmable way to correct the genetic defects underlying hereditary diseases. While CRISPR provides the programmable molecular scissors required to execute precise genomic modifications, deploying this technology safely and efficiently at a clinical scale presents monumental challenges, notably in predicting off-target effects, designing optimal guide RNAs, and understanding complex cellular repair mechanisms. This is where artificial intelligence (AI) comes in. By leveraging machine learning algorithms and deep neural networks, AI can rapidly process vast genomic datasets to identify optimal target sites, predict the precise outcomes of DNA repair, and engineer safer, higher-fidelity CRISPR variants. Together, CRISPR and AI could overcome the current limitations of gene editing and make personalized therapies a reality. This paper examines the role of AI in genome editing with CRISPR towards an advancement in precision medicine. It provides an overview of CRISPR genome-editing technologies and their applications in personalized healthcare, discusses the major technical, ethical, and clinical challenges limiting current systems, and explores how AI is being used to enhance guide RNA (gRNA) design, improve editing specificity, optimize delivery systems, and develop new genome-editing tools. Finally, the review highlights future perspectives and emerging opportunities at the intersection of CRISPR and AI that are expected to shape the next generation of personalized medicine.

2 CRISPR's Role in Precision Medicine

Precision medicine provides the diagnostic framework by identifying a patient's unique genetic variations and disease susceptibility. CRISPR, on the other hand, provides the molecular mechanism to act upon these findings. While precision medicine was traditionally limited to matching patients with existing pharmaceutical drugs based on biomarkers (e.g., specific genetic signals, mutations, or biological traits), CRISPR transforms this approach into a programmable and personalized intervention directly targeting a

patient's DNA. In this section, we first discuss the origin of CRISPR and then how its internal mechanisms uniquely align with the goals of precision medicine.

2.1 History of CRISPR: Past and Present

CRISPR was first identified in 1987 when Japanese researcher Yoshizumi Ishino and his team at Osaka University noticed unusual, repeating DNA sequences in the genome of E. coli bacteria (Ishino et al., 1987). At the time, the functional role of these arrays remained entirely unknown. Researchers working in the fields of microbiology and food science began studying these mysterious arrays that are now known as *CRISPR*. Subsequent microbial research revealed a crucial clue: the short DNA sequences within CRISPR arrays matched the genetic material of invading viruses. This discovery strongly suggested that these sequence arrays, working alongside *CRISPR-associated* (Cas) proteins, serve as an adaptive immune system in microbes. When a bacterium survives a viral infection, it integrates a snippet of the viral DNA into its own genome. This stored element, known as a *spacer sequence*, allows the bacterium to recognize and destroy the virus if it attacks again. Specifically, this DNA spacer sequence is transcribed into a gRNA, which directs Cas proteins to recognize and cleave matching viral genetic material upon reinfection. Of the many proteins encoded within bacterial CRISPR systems, Cas9 proved to be the key that unlocked CRISPR's use as a genome-editing tool. Naturally, Cas9 works alongside two RNA molecules to recognize and cut the DNA of invading viruses. This process is guided by complementary base pairing between the RNA and the target sequence. A cleavage occurs only if a short flanking sequence, called a protospacer adjacent motif (PAM), sits immediately next to the target site. Further research demonstrated that this RNA-programmed DNA-cleaving mechanism could be engineered to alter DNA sequences in virtually any cell type (Zheng et al., 2024). Over the past decade, CRISPR has evolved from a fundamental biological discovery into a multi-faceted technology used worldwide for functional genetic screens, reversible gene expression control, advanced genome editing, and molecular diagnostics across human cells, animals, and plants. Its therapeutic potential has transitioned rapidly into the clinic, laying the foundation for the active human clinical trials discussed later.

2.2 The Molecular Machinery and Human Adaptation

The classic CRISPR-Cas9 system consists of two primary components: the *Cas9 endonuclease* and a gRNA. Cas9 acts as programmable molecular scissors, using two distinct catalytic domains to cut both strands of DNA. The gRNA is a synthetic fusion of two bacterial components: a targeting CRISPR RNA (crRNA) and a structural trans-activating CRISPR RNA (tracrRNA). The crRNA contains a specific 20-nucleotide sequence that directs the complex, while the tracrRNA binds securely to the Cas9 protein. To execute a cut, the Cas9-gRNA complex first scans the genome to locate a PAM, and then the gRNA unzips the DNA double helix to test for a perfect match. If the sequence aligns flawlessly, Cas9 undergoes a structural shift and cleaves the target DNA. Adapting this microbial defense system for human cells required overcoming significant biological barriers. In 2012, researchers also demonstrated that the crRNA and tracrRNA could be fused into a single, streamlined molecule called a *single-guide RNA* (sgRNA), making the system far

easier to deliver. CRISPR-Cas9-mediated editing efficiencies reached 80% or higher at some target sites comparable to the previous techniques (Jinek et al., 2012). Just months after Jinek et al.'s 2012 discovery, three independent studies published in January 2013 demonstrated that this RNA-guided system could edit the genomes of human cells including human embryonic kidney cells, leukemia cells, and induced pluripotent stem cells (Doudna and Charpentier, 2014). Researchers quickly began using this technology to model human diseases by introducing paired double-strand breaks (DBS) at defined genomic positions. By delivering this engineered sgRNA and human-optimized Cas9 into mammalian systems, scientists unlocked the ability to precisely edit human DNA setting the stage for modern precision medicine therapies. Although CRISPR enables precise DNA editing, the efficacy of an edit depends on selecting the right target, avoiding errors, and building an optimized genome editor. These steps require searching through a vast combinatorial space to find the optimal setup.

3 CRISPR-Based Genome Editing Techniques

The core idea of genome editing is to utilize molecular tools to enter a target cell, locate a precise DNA site, and introduce specific structural changes to the genetic sequence. There are five categories of CRISPR-based genome editing tools. The original *nuclease editing* cuts both strands of the DNA and relies on the cell's natural repair pathways to either rejoin the broken ends directly or use matching sequences for alignment. This process deletes targeted genome areas, thus making it ideal for knocking out a gene. The *base editing* method chemically swaps one nucleotide base for another without cutting both strands, thus making it a safer choice to fix single-nucleotide mutations. *Prime editing*, on the other hand, is a search-and-replace technique that allows writing small changes using a supplied template. *Transposases and recombinases* extend this idea to larger DNA payloads where a guide-directed Cas protein locates the target site and recruits a separate transposase or recombinase enzyme to physically insert, delete, or rearrange a larger segment of DNA. Lastly, *epigenome editing* regulates gene expression by chemically modifying the surrounding proteins to turn genes on or off. Therefore, genome editing efficacy primarily depends on target *accuracy*, editing *precision*, sequence insertion *mechanics*, and efficient biological *delivery* vehicles. We expand on these methods below to further understand their trade-offs in applied precision medicine.

3.1 Nuclease Editing

As introduced above, nuclease editing exploits Cas9's native PAM-dependent cleavage mechanism to generate a DSB at the target site. This DSB is naturally resolved by non-homologous end joining (NHEJ) or microhomology-mediated end joining (MMEJ), which introduces random insertions or deletions (indels) at the cut site that can be harnessed for gene disruption (knockout). The Cas9 family of enzymes is most often used for genome editing applications. Meanwhile, Cas12 enzymes target T-rich PAMs to generate staggered DSBs; while wild-type Cas12 variants initially exhibited lower editing efficiency than Cas9, engineered Cas12a variants demonstrate comparable performance. Additionally, Cas13 enzymes target RNA instead of DNA; because RNA transcripts are transient, Cas13 modifies gene expression without permanently altering the host genome. However, the reliance on double-stranded breaks limits the utility of traditional CRISPR-

Cas nucleases for applications requiring precise correction of disease-causing mutations, as DSBs can trigger severe cellular toxicity, DNA damage responses, and large-scale chromosomal rearrangements. This limitation directly motivates the development of gene editing systems that function in a DSB-independent manner (Thomson et al., 2026).

3.2 Base Editing

Base editors (BEs) use RNA-programmable CRISPR-Cas proteins to modify individual DNA bases within a specific targeting window relative to the target protospacer. Consequently, the presence and positioning of a suitable PAM sequence are critical, as they determine whether the target base falls within the active catalytic window (Anzalone et al., 2020). BEs introduce targeted point mutations without generating harmful double-stranded breaks (DSBs) or requiring a homology-directed repair (HDR) template, thereby enabling precise genome editing even in HDR-deficient cells. BEs function by fusing a deaminase enzyme to a nuclease-impaired or catalytically inactive Cas protein. Once the single-guide RNA (sgRNA) directs the Cas complex to the target site, R-loop formation exposes a single-stranded DNA segment, allowing the tethered deaminase to chemically modify specific bases. For example, within a narrow targeting window, cytosine base editors (CBEs) convert C·G to T·A base pairs, whereas adenine base editors (ABEs) convert A·T to G·C base pairs (Pacesa et al., 2024). Because BEs produce targeted alterations with high predictability, they serve as powerful tools for disease modeling and hold significant therapeutic potential for correcting single-nucleotide mutations. However, key technical challenges remain, including bystander editing within the activity window, off-target genomic modifications, and unintended indels (Pacesa et al., 2024).

3.3 Prime Editing

Prime editing (PE) is a versatile gene-editing technology capable of executing precise base substitutions as well as targeted insertions and deletions (indels). It functions by fusing a Cas9 nickase to a reverse transcriptase (RT) enzyme and relies on a specialized prime editing guide RNA (pegRNA). The pegRNA directs the complex to the target genomic site and serves as the template for reverse transcription. Due to this versatility, PE can theoretically correct over 90% of known human pathogenic genetic variants while exhibiting significantly lower off-target activity than traditional nucleases. However, its clinical translation is constrained by lower editing efficiency and the requirement for target-specific optimization (Thomson et al., 2026). Recent efforts focus on optimizing PE architecture and pegRNA design to overcome these limitations.

3.4 Transposases and Recombinases

Traditionally, inserting new DNA sequences has relied on HDR. However, HDR only functions in actively dividing cells. This restriction rules out many mature human cells that scientists need to edit. Transposase and recombinase technologies offer a way around this barrier. Here, a non-cleaving Cas protein acts strictly as a molecular GPS. It uses its guide RNA to locate the target sequence without cutting the strands. Separately, transposase proteins bind to a flanking-sequence-labeled cargo DNA payload. The transposase physically

inserts this cargo directly at the target genomic address. Upon integration, it leaves short duplicated sequence on both sides (Anzalone et al., 2020). A related strategy fuses a non-cutting Cas9 directly to a recombinase enzyme. This engineered fusion system can precisely insert, delete, or invert DNA segments. Researchers have successfully used this programmable tool to delete DNA inside human cells. However, human cellular efficiency remains low because the recombinase is highly sequence-selective. Consequently, while highly reliable in bacteria, this approach is still in its infancy for human therapies.

3.5 Epigenome editing

CRISPR technologies can not only edit the genome but also regulate gene expression without altering the underlying DNA sequence. CRISPR-based epigenome editing uses a dCas9 fused to epigenetic-modifying domains to control gene expression at a chosen locus. CRISPRa (activation) and CRISPRi (inhibition) successfully turn genes on or off, but their effects are transient, reverting once the fusion protein is no longer present in the cell. To achieve durable modifications, researchers developed CRISPRoff, which attaches a KRAB repressor and a DNA methyltransferase to induce long-term gene silencing, and CRISPRon, which reverses this silencing by fusing a DNA demethylase alongside additional transcriptional activator proteins. Together, these tools allow gene expression to be programmed, silenced, and reactivated in a controlled, reversible manner (Thomson et al., 2026).

4 Current Clinical Applications in Precision Medicine

CRISPR-based therapies have demonstrated significant potential to transform precision medicine by enabling targeted modification of disease-associated genes. A landmark example is Victoria Gray, whose successful treatment with an early CRISPR-based therapy for sickle cell disease demonstrated the potential of gene editing to provide sustained clinical benefits for patients with severe genetic disorders (Brice, 2026). This progress gets the FDA approval of Casgevy (exagamglogene autotemcel) in December 2023, the first FDA-approved CRISPR-based therapy for eligible patients with sickle cell disease (U.S. Food and Drug Administration, 2023), followed by its approval for transfusion-dependent β-thalassemia in 2024, and a further expansion in 2026 extending eligibility to patients as young as 2 years old with either condition—making Casgevy the first approved genetic therapy indicated for children this young for both diseases (U.S. Food and Drug Administration, 2026). Casgevy uses CRISPR-Cas9 to edit patients' own hematopoietic stem cells to increase fetal hemoglobin production and reduce complications associated with defective adult hemoglobin. Another important development is the first-in-human CRISPR-Cas9 gene therapy trial for HIV (Richard K., 2024), in which EBT-101 was delivered using an adeno-associated virus 9 (AAV9) vector to target latent HIV proviral DNA. The early-phase study demonstrated targeted editing with no identified off-target DNA damage so far and showed a substantial reduction in the HIV reservoir in one participant.

Other notable developments, including the personalized treatment of Baby KJ for a rare genetic disorder, further demonstrate the potential of CRISPR-based therapies (Children's Hospital of Philadelphia (CHOP), 2025). More broadly, CRISPR-based approaches are being investigated in a growing pipeline of clinical trials

spanning cancer, muscular dystrophy, type 1 diabetes, cardiovascular disease, infectious disease, autoimmune disorders, and other inherited genetic disorders. According to CRISPR Medicine News (2026a,b), 285 trials are currently ongoing across this broad range of disease categories and editing strategies, with blood cancers constituting the largest single group (Innovative Genomics Institute, 2026). Collectively, these developments demonstrate the transition of CRISPR from an experimental gene-editing technology toward clinically applicable and increasingly personalized therapies, although challenges involving delivery, off-target effects, long-term safety, immune responses, and equitable access remain important considerations.

5 Technical Limitations of Current CRISPR Editing Techniques

Nuclease-based editing techniques, such as Cas9 and Cas12a, are powerful genome-editing tools but face several limitations. Off-target activity remains a major challenge, as unintended DNA cleavage can lead to unwanted genetic changes. In addition, the NGG PAM requirement of Cas9 and the T-rich PAM requirement of Cas12a restrict the range of genomic sites that can be targeted. DNA repair following nuclease-induced DBS can also produce heterogeneous insertions and deletions, resulting in unpredictable editing outcomes. Although engineered sgRNAs can improve targeting specificity and reduce off-target activity, delivery remains a significant barrier. The relatively large size of Cas9 and its guide RNA can make delivery into certain cells and tissues difficult, particularly for in vivo therapeutic applications. While some Cas12a variants are smaller than SpCas9, efficient and targeted delivery of the nuclease and guide RNA remains a major challenge (Anzalone et al., 2020).

BEs face several limitations that can affect their precision and therapeutic potential. Bystander editing can occur when multiple editable bases fall within the editing window, causing unintended changes alongside the desired edit. BEs are also mainly limited to transition mutations, meaning they cannot directly introduce transversions such as C→G or C→A. In addition, PAM availability and editing-window positioning can restrict which genomic sites can be targeted. RNA off-target editing is an important concern because some deaminase enzymes can naturally act on RNA molecules. Finally, there is a trade-off between editing-window size and bystander risk: wide-window editors reach more targets but risk more collateral edits; narrow-window editors are cleaner but reach fewer targets (Anzalone et al., 2020).

Despite this versatility, prime editing has not yet reached the clinical maturity of CRISPR nucleases and base editors. Editing efficiency varies considerably depending on cell type and cell state for reasons that remain poorly understood. It is still unclear exactly which patterns of local DNA repair activity lead to a successful edit rather than an unproductive one. Delivery is a separate obstacle because both the prime editor protein and its pegRNA are relatively large. Therefore, packaging and delivering them efficiently into target tissues in vivo remains technically demanding. One promising engineering lever targets the repair step directly. After reverse transcription, the newly edited 3′ DNA flap must outcompete the original, unedited 5′ flap to be incorporated into the genome. The cell must then be guided to resynthesize the remaining unedited strand using the newly edited strand as its template. Biasing repair toward both outcomes has been shown to increase editing efficiency and make the technique compatible with a wider range of cell types. Smaller, more efficient reverse transcriptase variants would likewise simplify packaging prime editors for in vivo

delivery. As these mechanistic and engineering challenges are resolved, prime editing is expected to become viable across an increasingly wide range of research and clinical applications (Anzalone et al., 2020).

Currently, the integration of AI with CRISPR follows three distinct dimensions. First, AI is used to find the best gRNA for existing genome editors to steer CRISPR precisely to its target. Second, AI helps redesign and improve the editors themselves by predicting protein structures more accurately, optimizing protein formation, and scaling the overall editor evaluation workflow. Third and finally, AI is helping scientists discover entirely new CRISPR systems and engineer synthetic ones from the ground up.

6 Role of Artificial Intelligence in CRISPR

AI and machine learning help overcome several limitations of CRISPR and improve its application in precision medicine. AI-driven models can optimize guide RNA design, predict off-target effects, identify suitable genomic targets, and improve the accuracy of editing-outcome predictions. Machine learning approaches can integrate large-scale genomic, transcriptomic, and clinical datasets to better understand genetic variants, cellular responses, and DNA repair mechanisms, enabling more personalized CRISPR treatment strategies. By combining computational prediction with experimental validation, AI has the potential to make CRISPR-based precision medicine safer, more efficient, and more accessible for a wider range of genetic diseases (Thomson et al., 2026).

6.1 AI for Guide RNA Design

A poorly designed sgRNA produces weak editing efficiency and increased off-target activity. So, selecting the optimal guide before laboratory testing is a critical first step. AI models predict gRNA binding efficiency, Cas9 cutting activity, and the sequence features that drive editing success, enabling faster and more reliable experimental design. DeepCpf1, a pioneering convolutional neural network (CNN), predicts sgRNA activity for Cas12a using data from a screen of over 15,000 sgRNAs in HEK293T and HCT116 cells, while the related DeepSpCas9 model achieves similarly strong predictive accuracy for Cas9. CRISPR_HNN subsequently improves prediction accuracy by capturing position-specific base substitutions within the spacer sequence. PLM-CRISPR incorporates a Cas9 variant's amino acid sequence via the ESM-2 protein language model to generalize predictions to variants lacking training data. DeepCRISTL applies transfer learning to fine-tune the CRISPRon model on smaller, cell-context-specific datasets. TnpB editing efficiency predictor (TEEP) extends this prediction task beyond Cas enzymes and forecasts the nuclease activity of TnpB-guiding ωRNAs (Thomson et al., 2026).

6.2 AI for Off-Target Prediction

Beyond on-target efficiency, prediction models help eliminate candidate gRNAs with a higher likelihood of off-target editing before they are ever tested experimentally. Deep learning models such as DeepCRISPR and CnnCRISPR predict off-target efficiency directly. CRISOT incorporates molecular dynamics simulations into

a machine learning framework and demonstrates up to 2-fold improved performance over earlier methods. crispAI advances this further by providing uncertainty estimates rather than a single-point score. These provide researchers a more informed view of off-target risk. Similar approaches extend to RNA-targeting systems as well. DeepCas13 and TIGER predict on- and off-target activity for Cas13d, while CRISPR-Embedding and CCLMoff incorporate k-mer clustering and RNA language models, respectively, to improve off-target prediction accuracy across diverse test datasets (Thomson et al., 2026).

6.3 AI for Predicting Editing Outcomes

Beyond predicting whether a gRNA will edit efficiently, AI models increasingly predict what the resulting edit will actually look like. For nuclease editing, inDelphi, Lindel, FORECasT, and SPROUT all forecast the specific DNA sequence produced following double-strand break repair based on a range of statistical approaches. inDelphi combines linear regression with neural networks, Lindel and FORECasT use logistic regression, and SPROUT applies gradient boosting. These capabilities allow researchers to exploit predicted microhomology pathways for more precisely targeted sequence integration. Because base editing requires strict alignment between the editing window and target base, a parallel set of models has emerged for this modality. BE-Hive, DeepABE/DeepCBE, and BE-DICT predict on-target efficiency and bystander effects; CGBE-Hive extends these predictions to C-to-G transversions; DeepBE recommends the optimal Cas9 PAM variant for a given target; BEguider extends prediction to near-PAM-less editors; and ABEdeepoff and CBE-deepoff forecast base-editor-specific off-target risk. Prime editing, which requires simultaneously optimising a pegRNA's spacer, primer binding site, and reverse transcriptase template lengths, has followed a similar trajectory. Early models such as DeepPE and EasyPrime addressed simple base substitutions, PRIDICT and DeepPrime scaled training data to predict a wider range of pathogenic mutations, and PRIDICT2.0, ePRIDICT, and MinsePIE further expanded prediction to complex edits, multiple base replacements, and chromatin-context effects. Most recently, OPED (a nucleotide language model) and DTMP-Prime (a deep transformer model) incorporate RNA secondary structure and pegRNA–DNA interactions to outperform earlier prime editing prediction models. Prediction tools for epigenome editing, such as EpiCas-DL and launch-dCas9, remain comparatively early-stage, reflecting the added complexity of context-dependent epigenetic regulation relative to direct DNA editing (Thomson et al., 2026).

6.4 AI for Personalized Precision Medicine

The integration of CRISPR screening with single-cell profiling technologies has generated datasets large enough to support AI-powered "virtual cell" models capable of predicting a cell's response to a genetic intervention *in-silico* before it is ever performed experimentally. Early efforts such as GEARS and PerturbNet used bespoke frameworks to forecast transcriptomic changes from unseen perturbations. The focus has since shifted toward large-scale foundation models including scGPT, Geneformer, and STATE. These models are trained on large-scale single-cell profiles and capable of generalizing predictions across unseen genes and cell types. The approach further extends with tools like Celcomen which integrates spatial transcriptomics with CRISPR perturbation data to model not only a cell's own response but also its effects on surrounding

tissue organization and intercellular communication. This capability is directly relevant to selecting patient- and tissue-specific editing strategies (Thomson et al., 2026).

AI's contribution to CRISPR is nonetheless bounded by the data it is trained on. CRISPR datasets can be limited, heterogeneous, and biased toward particular cell types, organisms, or experimental conditions. This reduces how well AI models generalize across different populations and clinical settings. Incomplete variant annotation and underrepresented populations in training data may further contribute to algorithmic bias. AI predictions may also fail to fully capture the complexity of cellular environments, including chromatin accessibility, DNA repair pathways, and cell-specific responses. The interpretability of complex machine-learning models also remains an important concern when predictions direct therapeutic decisions (Thomson et al., 2026). The use of AI should therefore be viewed as a supporting tool rather than a drop-in replacement for experimental validation and clinical expertise.

7 Challenges and Ethical Concerns

CRISPR gene editing has opened new possibilities for treating complex genetic diseases, but its increasing success also raises significant ethical, social, and safety concerns. Somatic editing, which is limited to an individual and not inherited by future generations, is generally considered more ethically acceptable when appropriate safety, consent, and regulatory requirements are met—and it is precisely why the therapies discussed above (Casgevy, the EBT-101 HIV trial, and Baby KJ's treatment) have advanced to clinical approval more readily than germline approaches would. Germline editing, by contrast, modifies embryos, sperm, or eggs, meaning changes can be inherited by individuals who cannot consent to them. In 2018, Chinese researcher He Jiankui announced the birth of genetically edited twins whose embryos had been altered using CRISPR-Cas9 to modify the CCR5 gene, an experiment that drew widespread international criticism over inadequate consent, scientific justification, and the heritable nature of the changes (Raposo, 2019). The case also intensified debate over genetic enhancement—whether editing used to prevent disease should also be permitted to influence traits such as appearance, athletic ability, or cognition. Nobel laureate Emmanuelle Charpentier was among the signatories of a widely cited call for a moratorium on heritable genome editing, underscoring the need for internationally coordinated governance over which applications should be permitted, restricted, or prohibited (Lander et al., 2019).

A further challenge is the high cost and limited accessibility of CRISPR-based therapies. Personalized gene-editing treatments require sophisticated manufacturing, specialized clinical infrastructure, and extensive quality control and monitoring—reflected in list prices such as Casgevy's $2.2 million per patient (Rueda et al., 2024). Such costs risk creating substantial healthcare inequality. Advanced therapies remain accessible mainly to well-insured patients in high-income countries. Many low- and middle-income healthcare systems, meanwhile, lack the genomic testing, cell-processing, and specialist infrastructure required to deliver them at all. Improving affordability, scalability, and global availability will therefore be essential if CRISPR-based precision medicine is to benefit a broad population rather than a privileged few. Balancing CRISPR's promise against its ethical risks and economic barriers will require careful, ongoing societal deliberation alongside continued scientific progress.

8 Conclusion and Future Perspectives

CRISPR has evolved from a bacterial adaptive immune mechanism into one of the most powerful technologies for genome engineering and has already demonstrated significant clinical potential, particularly in the treatment of genetic diseases such as sickle cell disease and β-thalassemia. The development of nucleases, BEs, prime editors, and CRISPR-associated transposases has progressively expanded the precision, versatility, and potential applications of genome editing. However, challenges including off-target effects, delivery, unpredictable DNA repair, limited editing windows, large-scale genomic data limitations, AI bias and interpretability, ethical concerns, and high treatment costs continue to restrict widespread clinical implementation. AI offers an important opportunity to address several of these challenges by enabling more accurate prediction, optimisation, and personalisation of CRISPR-based editing. Therefore, the convergence of CRISPR, AI, and precision medicine represents a promising pathway toward safer, more targeted, and personalised therapies. Realising this potential will depend not only on technological innovation but also on ethical governance, rigorous clinical validation, responsible use of genomic data, affordability, and equitable global access, ensuring that the benefits of genome editing extend beyond technological achievement to meaningful improvements in human health.

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